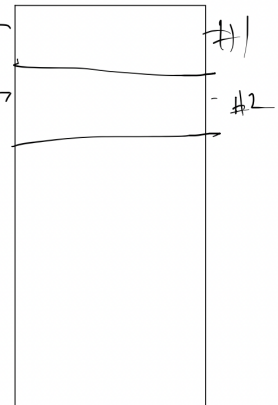


```
export default function App() {
  let [name, setName] = useState<string>("");
  let [age, setAge] = useState<number>(18);
}
```

Tuple

HEAP



```
export default function App() {
  let [name, setName] = useState<string>("");
  let [age, setAge] = useState<number>(18);
}
```

age = age + 10

Change the state

NO Reconciliation

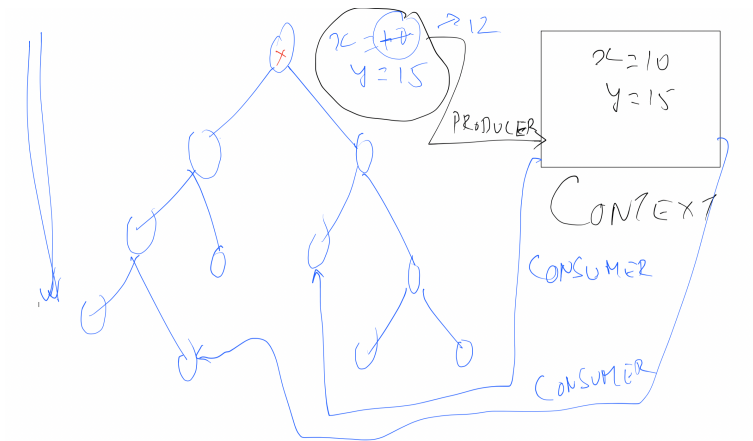
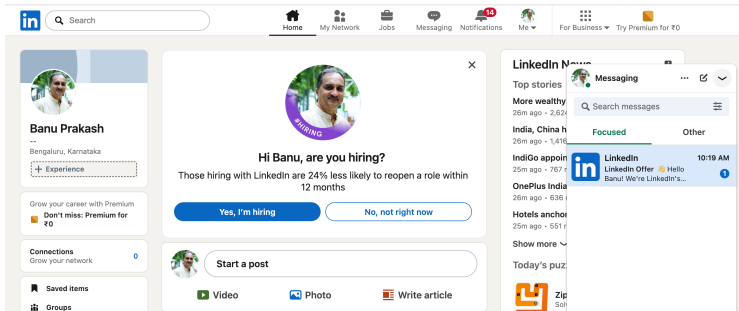
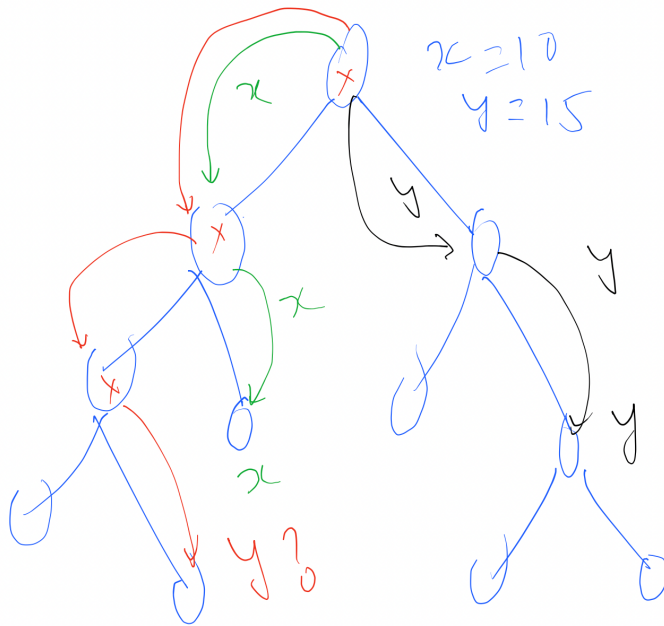
```
export default class App extends Component {
  state = {
    name: string = "",
    age: number = 18
  }

  setName(name) {
    this.setState({
      name
    })
  }

  setAge(age) {
    this.setState({
      age
    })
  }
}
```

FORCE
RENDER

NO ReRender



```

export default function cartReducer(state: CartStateType, action : Action) {
  switch(action.type) {
    case 'ADD_TO_CART':
      const product = {...action.payload}
      let item :CartItem = {...product, qty: 1, amount: product.price}
      // return new object
      // no usage of state.items.push(item)
      return {
        items: [...state.items, item],
        quantity : state.quantity + 1,
        total: state.total + item.amount
      }
    case 'INCREMENT':
      return state;
    case 'CLEAR CART':
      return state;
    default:
      return state;
  }
}

```

```

const initialState: CartStateType = {
  items: [],
  quantity: 0,
  total: 0
}

```

```

let [state, dispatch] = useReducer(cartReducer, initialState);

```

```

items: [],
quantity: 0,
total: 0

```

```

export default function cartReducer(state: CartStateType, action : Action) {
  switch(action.type) {
    case 'ADD_TO_CART':
      const product = {...action.payload}
      let item :CartItem = {...product, qty: 1, amount: product.price}
      // return new object
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      return {
        items: [...state.items, item],
        quantity : state.quantity + 1,
        total: state.total + item.amount
      }
    case 'INCREMENT':
      return state;
    case 'CLEAR CART':
      return state;
    default:
      return state;
  }
}

```

```

const initialState: CartStateType = {
  items: [],
  quantity: 0,
  total: 0
}

```

```

let [state, dispatch] = useReducer(cartReducer, initialState);

```

```

items: [],
quantity: 0,
total: 0

```

```

dispatch({type: 'ADD_TO_CART', payload: ...})

```

Act m

